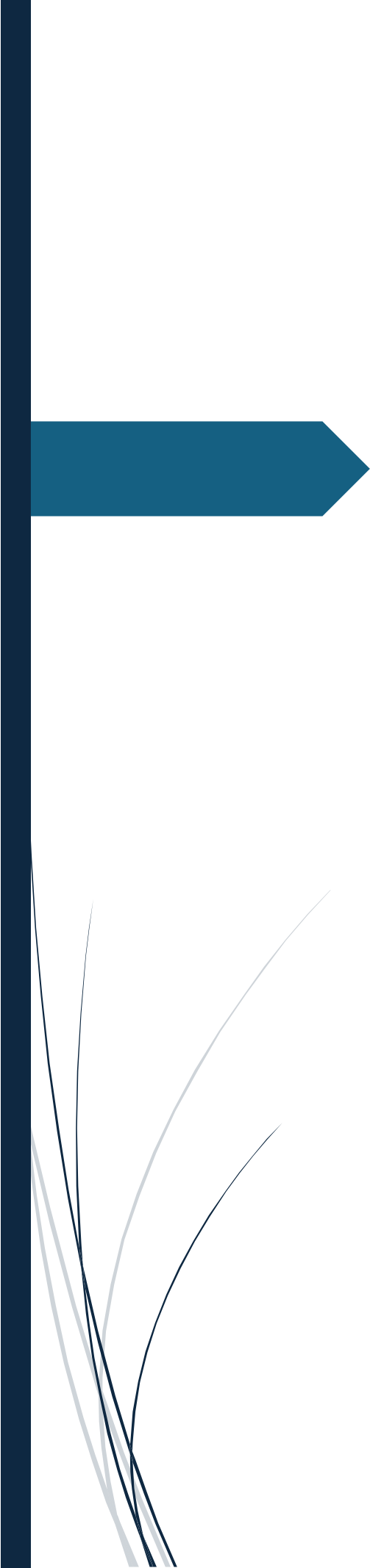
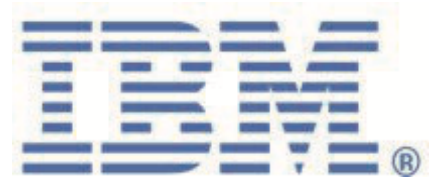


A decorative graphic on the left side of the page. It features a thick, dark blue vertical bar. A horizontal blue arrow points to the right from the middle of this bar. At the bottom of the vertical bar, there are several thin, curved, light blue lines that sweep upwards and to the right.

Maximo Monitor 9.1 Managed Gateway Modbus Lab

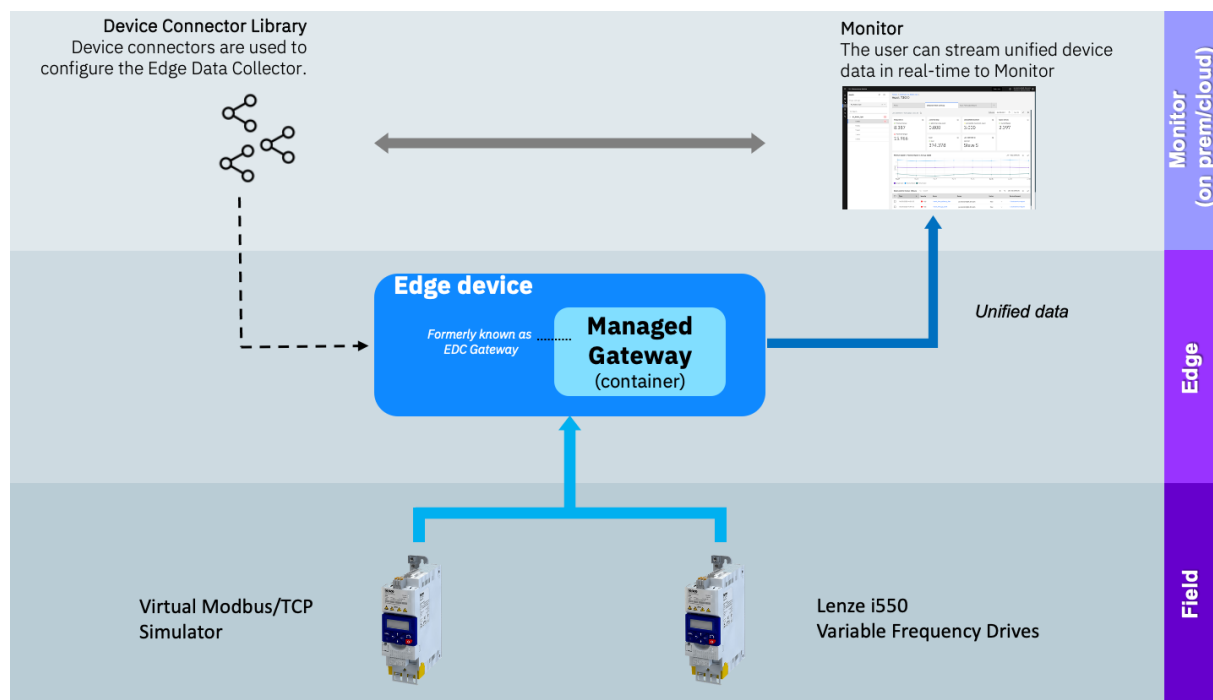
Table of Contents

Overview.....	3
Exercise 1. Setup Simulator Environment	4
Exercise 2. Create a Managed Gateway.....	6
Exercise 3. Add an Industrial Device	8
Exercise 4. Deploy and Verify Data Inflow	11
Exercise 5. Add another Industrial Device.....	15
Exercise 6. Deploy and View in Dashboard	17
End	

Overview

In this lab you will learn how to setup a Managed Gateway within Maximo Monitor 9.1 and adding a couple of OT devices from the Device Library.

You will learn the steps needed to successfully setup a Managed Gateway using some virtual simulated Variable Frequency Drives (VFD) that will deliver the device data over Modbus/TCP via the Managed Gateway into Maximo Monitor using the pre-configured connector for Lenze i550 VFD from the Device Connector Library.



1. In the provided URL, replace **admin** with **api**, then press **Enter**.
2. An error message will appear — this is expected.
3. Next, replace **api** back with **admin** in the URL and press **Enter** again.

Note: When switching between the MAS module and MAS Monitor, you may be prompted to proceed to a site without a CA-authority-signed certificate. Accept the warning to continue.

The exercises will cover:

- Setting up the simulator environment
- Create a Managed Gateway
- Add devices to the Manage Gateway
- Verify data flow from Modbus/TCP Variable Frequency Drives all the way into Maximo Monitor

Exercise 1. Setup Simulator Environment

In this Exercise you will learn how to: Install the Modbus simulator

1. You need the IP address of the machine where you run the simulator no matter which version you are using. You will need that later when configuring the Managed Gateway.

- **Get IP Address on macOS**

Use the following command: **ifconfig en0**

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether da:63:8e:d0:95:02
    inet6 fe80::1c86:af08:7369:24a1%en0 prefixlen 64 secured scopeid 0xe
    inet 192.168.1.3 netmask 0xffffffff0 broadcast 192.168.1.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active
```

- **Get IP Address on LINUX**

Use the following command: **hostname -I**

```
➔ ~ ssh pi@192.168.1.97
pi@192.168.1.97's password:
Linux RPE007 6.1.21-v7+ #1642 SMP Mon Apr 3 17:20:52 BST 2023 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jun 19 11:22:38 2024 from 192.168.1.42
pi@RPE007:~$ hostname -I
192.168.1.97 fde1:7a3d:7b2a:ef4e:200a:c9c5:dd61:a37e
pi@RPE007:~$
```

- **Get IP Address on Windows**

Use the following command: **ipconfig**


```

Command Prompt
Microsoft Windows [Version 10.0.19045.2364]
(c) Microsoft Corporation. All rights reserved.

C:\Users\014420678>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fde1:7a3d:7b2a:ef4e:370a:4b4b:3dc7:3393
    Temporary IPv6 Address. . . . . : fde1:7a3d:7b2a:ef4e:20fc:a178:fb5f:822a
    Link-local IPv6 Address . . . . . : fe80::c73a:4b4b:3dc7:3393%2
    IPv4 Address. . . . . : 192.168.1.43
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\014420678>

```

2. **Install Docker:** There are different ways to install the Docker engine depending of the package and the OS. One multi-platform option is **Rancher Desktop**.
3. It is fairly easy to install Rancher Desktop, as you just have to follow this guide: [Running Docker locally](#)
4. **Create the Docker container:** Open a terminal or command window and run the following command:

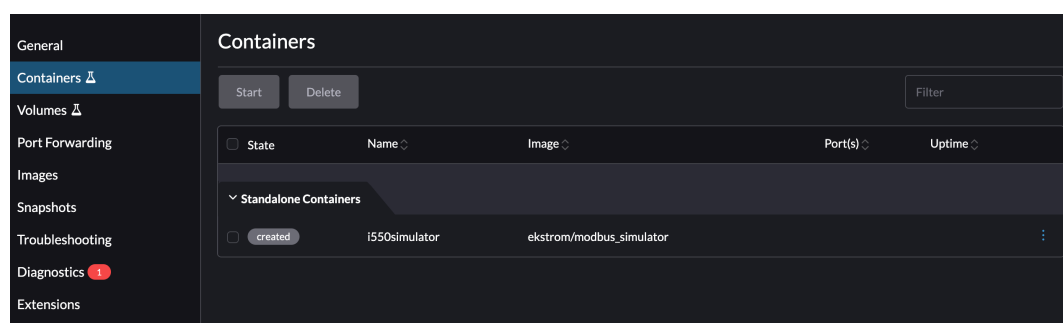
`docker create -p 10502:10502 -p 20502:20502 -v node_red_data_vol:/data --name i550simulator ekstrom/modbus_simulator`

```

tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker create -p 10502:10502 -p 20502:20502 -v node_red_data_vol:/data --name i550simulator ekstrom/modbus_simulator
8441c31b3dce75686bd79de6c9743031d14c72c39f12ac5fed10231f588dd0b7
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %

```

You can see a container got created in Rancher Desktop



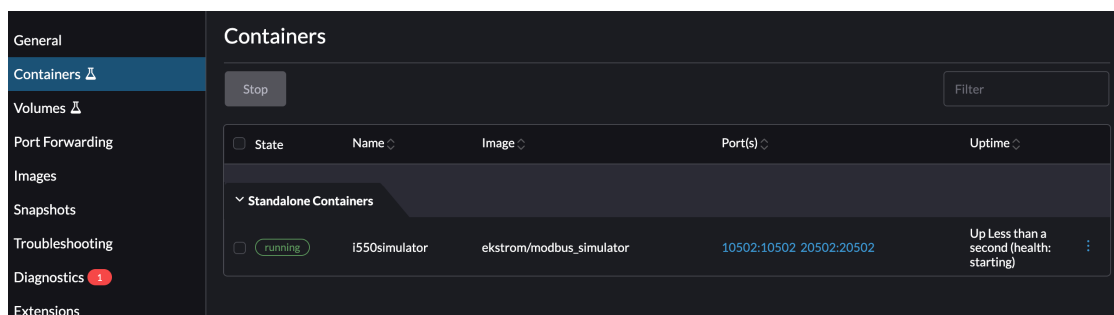
This command **sets up a Modbus simulator container (i550simulator)** that exposes ports 10502 and 20502 to the host and stores its persistent data in the node_red_data_vol volume. The container is only created.

5. **Start the Docker container:** Run the following command to start the container.

docker start i550simulator

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker start i550simulator
i550simulator
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %
```

You can see simulator is running in Rancher Desktop

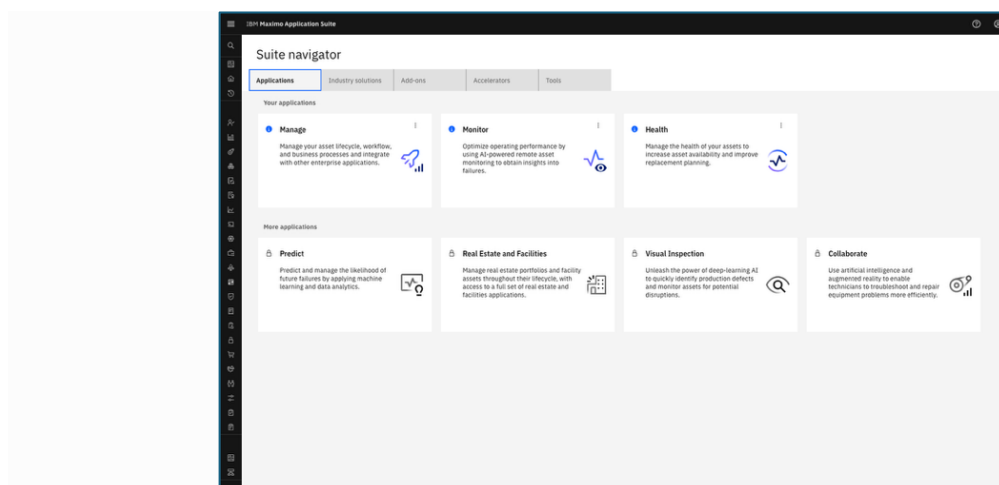


Note: The simulator is now active and the random and dynamic values will change every 30 second. It will run in the background and not produce any output in the terminal/command window.

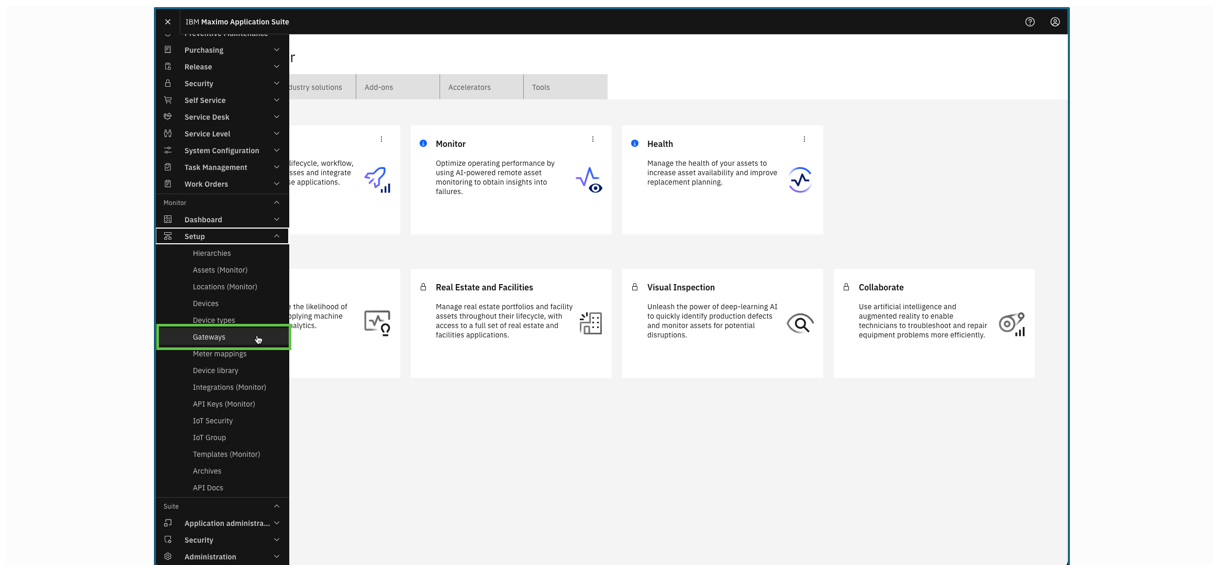
Exercise 2. Create a Managed Gateway

In this Exercise you will learn how to create the Managed Gateway in Monitor.

1. Login to Maximo Application Suite with your credentials.



2. Expand Setup under the Monitor section in the left menu and select **Gateways**:

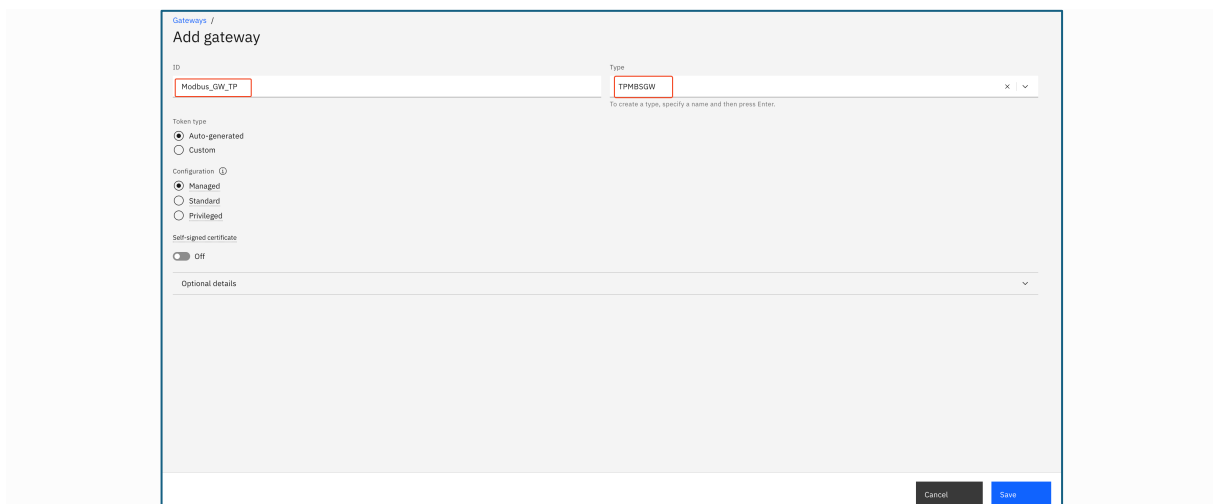


3. Select **Add Gateway**.

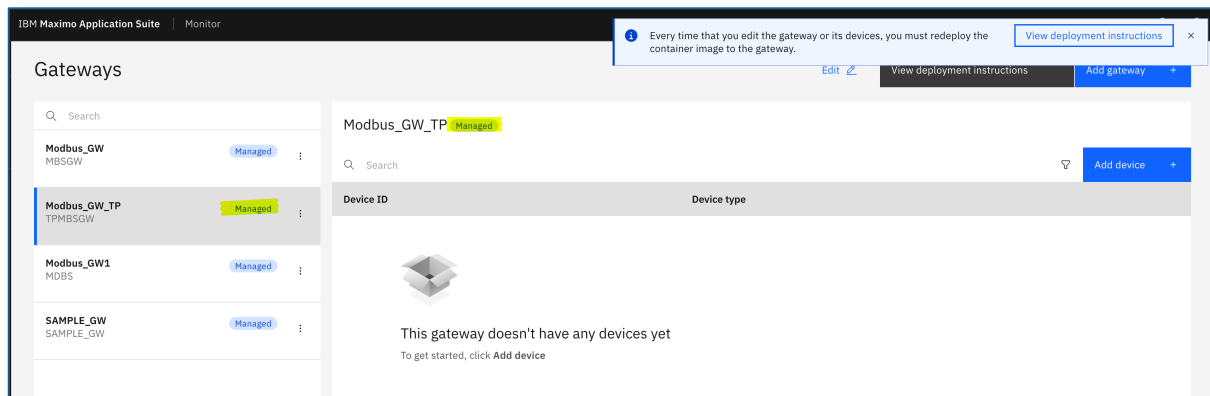


4. Provide gateway id and gateway type name. You can type preferred name and click enter.

- ID- **Modbus_GW_XX** (Replace XX with your initials)
- Type- **XXMBSGW** (Replace XX with your initials)
- Token Type- **Auto-generated**
- Configuration- **Managed**



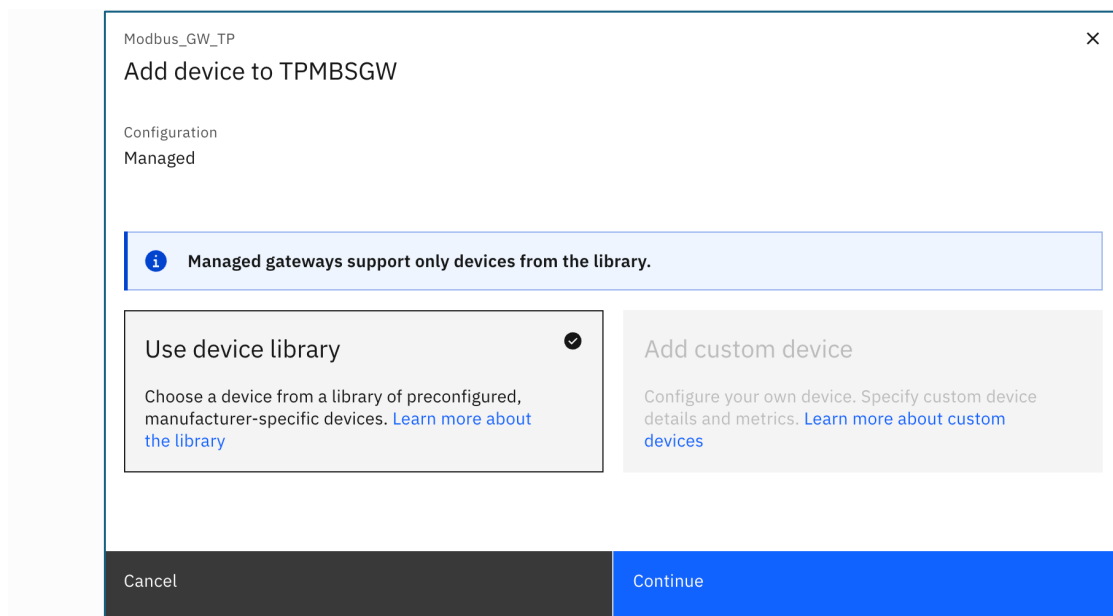
5. Click on **Save**.
6. You will now see your new Managed Gateway, including a **Managed** tag in both the list of Gateways as well as in the gateway definition.



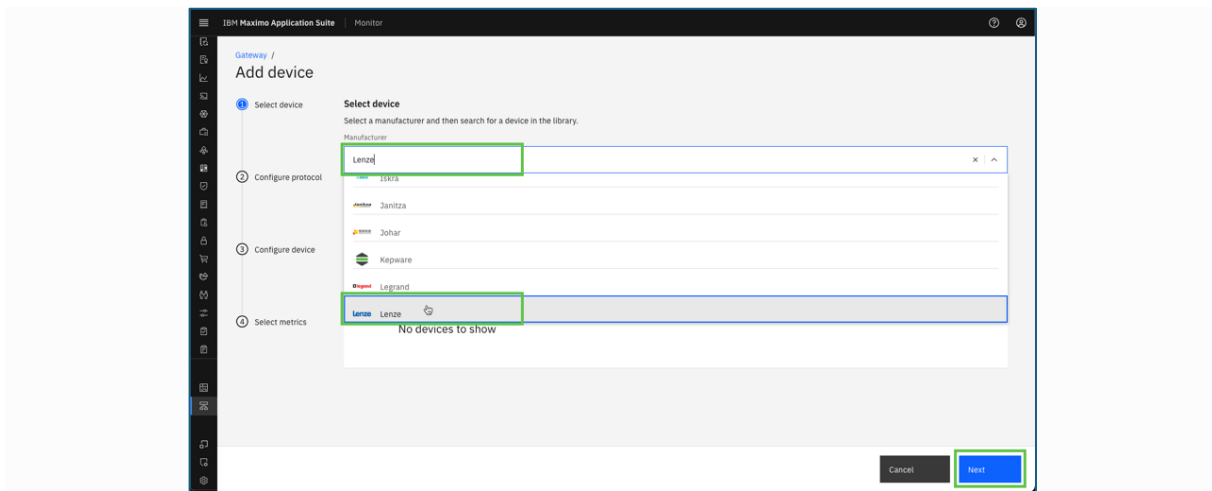
Exercise 3. Add an Industrial Device

In this Exercise you will learn how to add the first industrial device to the Managed Gateway.

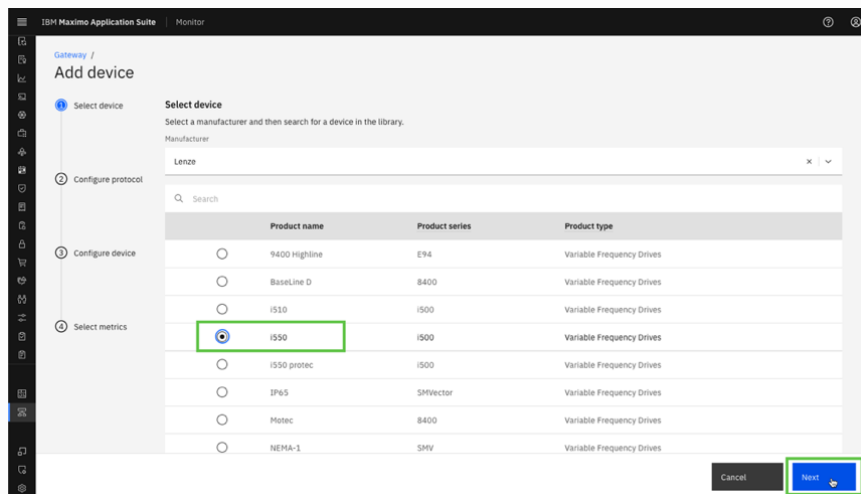
1. Click on **Add Device** and select **Use device library**. (Contd. From last exercise)



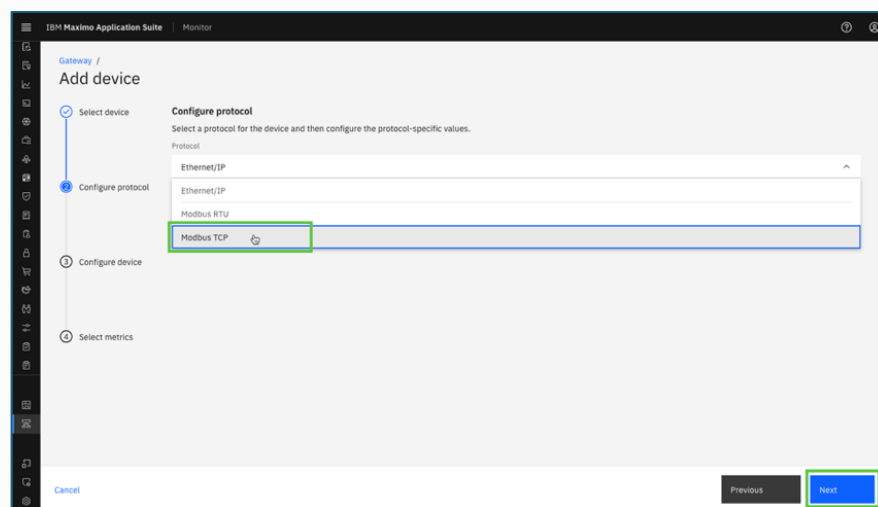
2. Search for **Lenze** in the manufacturer drop-down and select it. Click **Next**:



3. Select the **i550** product and click on **Next**.



4. Select the **Modbus TCP** protocol. Click on **Next**:



5. Now it is time to use the IP address of the simulator and combine it with port number 10502 separated by : , like **192.168.1.64:10502**. Click on **Next**.

Gateway / Add device

Select device

Configure protocol
Select a protocol for the device and then configure the protocol-specific values.

Protocol: **Modbus TCP**

IP address: **192.168.1.3:10502** Server or unit ID: **1**

Configure device

Select metrics

Summary

Cancel Previous Next

6. It will ask you to configure device and provide your own device id and device type. Click on **Next**.

- Device ID- **XX_Lenze_i550_01**
- Device type- **Lenze_DT_XX** (Replace XX with your initials)

Gateway / Add device

Select device

Configure device

Device ID: **TP_Lenze_i550_01**

Device type: **Lenze_DT_TP** Product type: **Variable Frequency Drives**

To create a type, specify a name and then press Enter.

Configure device

Select metrics

Summary

Cancel Previous Next

7. Select these three metrics and click on **Next**.

Gateway / Add device

Select metrics
Select metrics for the device and configure when data is sent. Selected metrics are added to the device as raw metrics. All of the metrics in the table are added to the selected device type.
[Learn more about metrics](#)

Data frequency in milliseconds: **1000** Change threshold (%): **On every change**

3 items selected

Metric	Unit	Data frequency (ms)	Change threshold (%)
☒ Control Card Temperature	degree Celsius	1000	On every change
☒ DC Bus Voltage	volt	1000	On every change
☒ Frequency Command	hertz	1000	On every change
☐ Motor Current	ampere	1000	On every change
☐ Motor Torque	percent	1000	On every change
☐ Motor Voltage	volt	1000	On every change
☐ Output Frequency	hertz	1000	On every change
☐ Torque Command	percent	1000	On every change
☐ Supply Voltage	volt	1000	On every change

Cancel Previous Next

8. Click on **Next** -> **Save**.

Gateway / Add device

Summary
You're almost done! Here's a summary of the device configuration details for your confirmation.

Device Details

Manufacturer	Lenze	Product name	i550
Gateway ID	Modbus_GW_TP	Gateway type	TPMBSGW
Device ID	TP_Lenze_i550_01	Device type	Lenze_DT_TP
Protocol	modbus-tcp	IPPort/ServerId	192.168.1.3:10502/1

Cancel Previous Save

9. You have now successfully added the first OT device to your Managed Gateway.

IBM Maximo Application Suite | Monitor

Every time that you edit the gateway or its devices, you must redeploy the container image to the gateway. [View deployment instructions](#)

Gateways

Search

- Modbus_GW (MBSGW) Managed
- Modbus_GW_TP (TPMBSGW) Managed**
- Modbus_GW1 (MDBS) Managed
- SAMPLE_GW (SAMPLE_GW) Managed

Modbus_GW_TP Managed

Search

Device ID	Device type
TP_Lenze_i550_01	Lenze_DT_TP

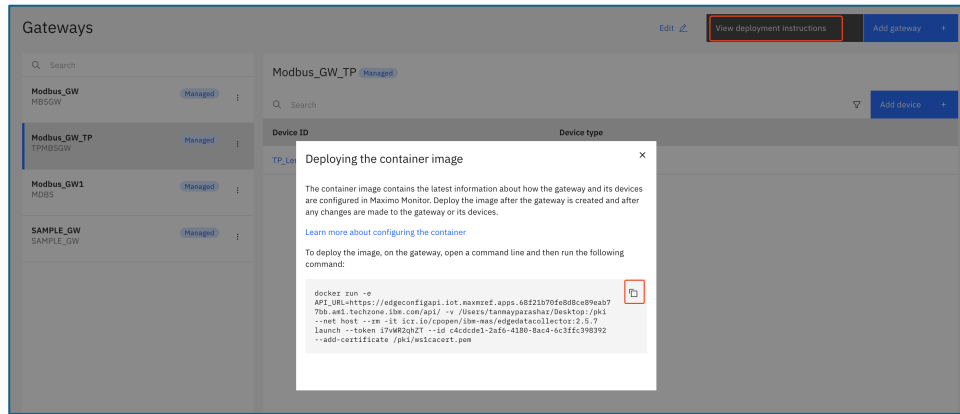
Add device +

Exercise 4. Deploy and verify data inflow

In this Exercise you will learn how to:

- Deploy the Managed Gateway
- Verify data inflow

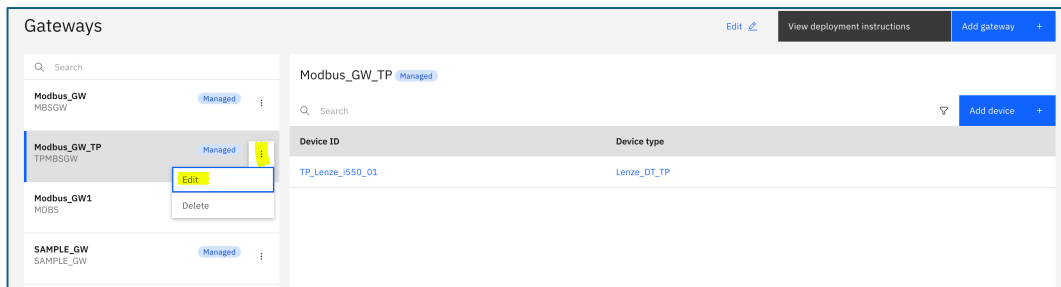
1. While looking at your Managed Gateway in the Gateways list, press the **View deployment instructions**. Click on the docker command to copy it to the clipboard.



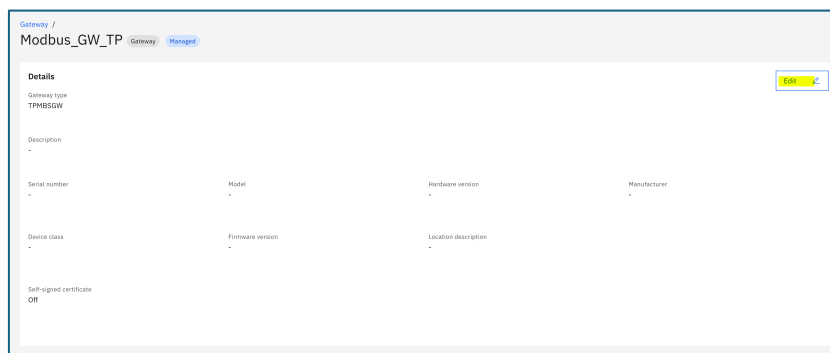
- Once you run this docker command in Terminal/Command Prompt you will encounter the below error.

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker run --s API_URL=https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/ --net host --rm -it icr.io/cpopen/ib
m-mas/edgedatacollector:2.5.7 launch --token i7vWR2qhZT --id c4cdcde1-2af6-4180-8ac4-6c3ffc398392
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-26T18:24:07.208824Z INFO: edge-cli/src/lib.rs:106: Arguments: launch --token i7vWR2qhZT --id c4cdcde1-2af6-4180-8ac4-6c3ffc398392
2025-10-26T18:24:07.211473Z INFO: edge-cli/src/lib.rs:108: Version: 2.5.7
2025-10-26T18:24:07.211592Z INFO: edge-cli/src/lib.rs:109: Git sha1: cabdca6f2bf101743db0f093cd83bb9763bc4668
2025-10-26T18:24:07.211692Z INFO: edge-cli/src/lib.rs:110: Branch: master
2025-10-26T18:24:07.211656Z INFO: edge-cli/src/lib.rs:111: Build number: 748
2025-10-26T18:24:07.211700Z INFO: edge-cli/src/lib.rs:112: Build ID: 41027330
2025-10-26T18:24:07.347811Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdc
e1-2af6-4180-8ac4-6c3ffc398392/
error sending request for url (https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdcde1-2af6-4180-8ac4-6c3ffc398392/)
Caused by:
0: client error (Connect)
1: error:1416f886:SSL routines:tls_process_server_certificate:certificate verify failed:ssl/statem/statem_clnt.c:1915: (unable to get local issuer certificate)
2: error:1416f886:SSL routines:tls_process_server_certificate:certificate verify failed:ssl/statem/statem_clnt.c:1915:
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %
```

- To fix the above error, go to Gateways you created in Monitor and click on Manage Gateway (three dots) and then click on **Edit**.



- Click on **Edit**.



- Enable **Self-signed certificate** and download the certificate from OCP and put the path of the certificate.

Edit Modbus_GW_TP

Description

Serial number

Model

Hardware version

Manufacturer

Device class

Firmware version

Location description

Self-signed certificate

On

Add certificate by using

Certificate authority path

Certificate and certificate private key path

Certificate authority path

/Users/tanmayparashar/Desktop/ws1cacert.pem

Download certificate

Click Download certificate, upload the certificate to the gateway, and then specify the path, for example: /home/gateway/certs/ca.crt

Cancel

Save

Note: To download the certificate from cluster navigate to **Workloads -> Secrets -> Project -> IoT -> Search public TLS -> Download ca.crt** and save the certificate with .pem extension.

Red Hat

OpenShift on IBM Techzone

Administrator

Home

Operators

Workloads

Pods

Deployments

DeploymentConfigs

StatefulSets

Secrets

ConfigMaps

Project mas-wsitaly-iot

Secrets

Filter

Name

wsitaly-public

wsitaly-public

Clear all filters

Name	Type	Size	Created
wsitaly-public-tls	kubernetes.io/tls	3	15 Oct 2025, 19:53

ca.crt

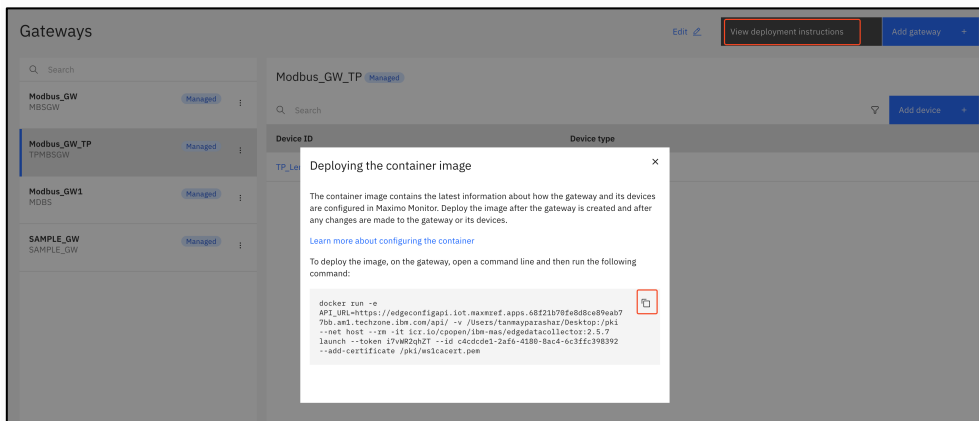
```

-----BEGIN CERTIFICATE-----
MIID1zCCA+gAwIBAgIRANRj+lf1ZDztUWgjdIV1IowDQYJKoZIhvcNAQELBQAw
gYQxCzAJBgNVBAYTAkdQDQYDQVQOQHEWZMb25kb24xZDANBgNVBAkTBkxvbmRv
b2EjMCEGA1UECzMLSUJUNIE1heG1tbyBBcHBSaWVibG1vZDZlbnR0ZSAoUHVibG1j
KTEjMCEGA1UECzMLSUJUNIE1heG1tbyBBcHBSaWVibG1vZDZlbnR0ZSAoUHVibG1j
MTM1MjQ2WhcNNDA0MDEwMTM1MjQ2WjCBhDELMAkGA1UEBHMCR0IzDzANBgNVBAcT
BkxvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRvbmRv
cGxpY2F0aW9uIFN1aXR1ChQdWJsaWwMjQ2WjCBhDELMAkGA1UEBHMCR0IzDzANBgNV
eS5tYXMuUWJtLnVbTCCASTwDQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBALzI
AQ/WVdGDMWdYDmLU1LHszm/JsUDGMjUkPuccRpf+1GVDBw09+m7N0YS0WY/RpyX5
oSaKcUHQToJkFacc/O0wCVAw8MDH7RuFw/ACChRVG4koaKvbJZbHgA6ca7zaTF9d8
qnbcTpdVitu0YnKxr102RM7J6K1jAmm0tXI0uVXB12Yjys0T7C0A4q2Z2qAndUwRr
E3BrjJgU4h8o+wxhJZFv2Kcca50x/t0k0o+H0opKU/e7mm2nsuW49yoxRczoucSL
TbJ4q26MLRmMtUd0b+0Z17jx5MXLLkuiERB91EUv/X/Ku10y11yWBSveMdBzcer6T
z7dWp1VWtudyFdv7f68CAwEAANCEAwDgYDVR0PAQH/BAQDAgTEMA8GA1UdEwEB
/wQFMAMBaf8wHQYDVR0BBYEF0BphpvT1+1Sn78BS05T1ptLMGujMA0GCSqGSIb3
DQEBcwAA4IBAQBjbx1PwAPKV3Q1KK619/3gxkRyS9cApD110hLD7P1cVaEV+vo
nLj12XV3nMS8XJTx6VKs9xvxxJWwDgSKNTf+fu8/NMhNHzXwr0k3kFGLBuScEj
wLX5WLAfIZnHag+JnqsYPc7cZT0w9YH4t8jbnHjMjpNkNa97FhcEK0qcpU09EJ6
s41wuE97j+EgTkm0JrYk/13+5uxbPqbqkSXfu1i+OwakiukPCZ45f+A408ePLwWe
7SyFGjg0J5uKGTJf6CAAdApNA0VTn0y6PzYFxdWJKJ03opvWEVKMzoAg1UH7qoyP5
pm739QAAMIMLL7gAZQ7sdxuzk40oD9ywaT0
-----END CERTIFICATE-----

```

```
Welcome  cacertmasmref.pem x
Users > tanmayparashar > Desktop > cacertmasmref.pem
1 -----BEGIN CERTIFICATE-----
2 MIID1zCCAR+gAwIBAgIRANRj+1f1ZDztUWgjdIV1IowDQYJKoZIhvcNAQELBQAw
3 gYQxCzAJBgNVBAYTAkQCMQ8wDQYDVQQHEwZMb25kb24xZDZANBgNVBAkTBkxvbmRv
4 bJEUcCwGA1UECzM1SUJNIEIheGltbYBBcHBsaWdhbG1vbmRvL0ZSAoUHVibG1j
5 KTEjMCEGA1UEAxAHh1bG1jLndzaXRhbHkubWZmLmlibS5jb20wHhcNMjUxMDE1
6 MTM1MjQ2WmcNNDUxMDEwMTM1MjQ2WjCBDELMAkGA1UEBhMCRC0xZDZANBgNVBAcT
7 BkxvbmRvbmRvZEPMA0GA1UECRMGTG9uZG9uZS4wLWYDVQQLYyVjQk0gTW4w1vIEFw
8 cGxpY2F0aW9uIFN1aXRlICQdWJsaWpMSMwIQYDVQQDExpWJsaWpMSMwIQYDVQDQ
9 eS5tYXMuWJtLmNvbTCCASIDQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBALzI
10 AQ/VWd6DMWdYdMLULHSzm/JsUDGMjUkPuccRp+LGVDBWQ9+m7N0YS0WY/RpyX5
11 oSaKcUhgToJKfac/00wCVAw8MDH7RuFw/ACChRV64koaKvbJZbHgA6ca7zaTF9d8
12 qnbcTpdVitu0YnKxr10ZRM7J6LjAmwDtXI0uVXBIZjysQTZCOA4q2ZqAndUwRr
13 E3BrjJgU4h8o+XhJZFv2Kcca5Gx/t0k0o+HDopKU/e7mm2nsuW49yoxRczouC5L
14 TbJ4q26MLRmMtuDgB+DZ17jx5MXLkuiERB9LEUv/X/Ku10y1lywBSvcMdSbzc6T
15 z7dWpLViudyFdv7f60CAwEAAANCEAwDgYDVR0PAQH/BAQDAgIEMA8GA1UdEwEB
16 /wQFMAMBAf8wHQYDVIR00BBEYEF0BphpvT1+1Sn78BS05T1ptLMGvuMA0GCSqGSIb3
17 DQEBwUAA4IBAQBjX1PwPKV3Q1KK619/3gXkRyS9cApD1i0hLdr7P1cVaEV+vo
18 nLj12XVv3nMS8XUJTx6KVs9xvKxJWdGSKNtF+fu8/NMhHNRKw0k3kFG1BuScEj
19 wLX5WLAf iZnHag+JngsYpC7cZT0w9YH4t8jbnHjMjpnKNa97FhcEK0qcpU09EJ6
20 s41wuE97j+EgTkm0JRyK/13+5uxbPqbkqSXFu1i+0wakiukPCZ4Sf+A400ePLwWe
21 7SyFGjg0J5kGTJf6CAApNA0VTn0y6PzYFxdWkJO3QpVwEVK04oAg1UH7qoyp5
22 pm739QAAmIML7gAQA7sdXuzk40oD9ywaT0
23 -----END CERTIFICATE-----
```

6. Click on **Save** and go to **Gateways** and copy the docker command by clicking on View deployment instructions.



7. Open a terminal window (Mac/Linux) or Command window (Windows) where you want to run the Managed Gateway and then paste the docker command line from the clipboard. Click **enter** to execute it, and you should see something similar to the following.

```
tanmayparashar@tanmay-MacBook-Pro-2: ~ % docker run -e API_URL=https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/ -v /Users/tanmayparashar/Desktop:/pk
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-26T18:38:13.883632Z INFO edge-cii/src/lib.rs:180: Arguments: launch --token 17wR2qhZT --id c4cdcd1-2af6-4188-8ac4-6c3ffc398392 --add-certificate /pki/wslcacert.pem
2025-10-26T18:38:13.886932Z INFO edge-cii/src/lib.rs:180: Version: 2.5.7
2025-10-26T18:38:13.886462Z INFO edge-cii/src/lib.rs:180: Git sha1: cab6ca6f2bf101743dbf093cd83bb9763bc4668
2025-10-26T18:38:13.886495Z INFO edge-cii/src/lib.rs:110: Branch: master
2025-10-26T18:38:13.886630Z INFO edge-cii/src/lib.rs:111: Build number: 748
2025-10-26T18:38:13.886657Z INFO edge-cii/src/lib.rs:112: Build ID: 4102738
2025-10-26T18:38:14.014018Z INFO /omniioth/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/installations/c4cdcd
e1-2af6-4188-8ac4-6c3ffc398392/
2025-10-26T18:38:15.317811Z INFO /omniioth/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/encrypted-profiles/
71d089410e-24a9-4a10-b21b-fe3da28dddb8&id=ef5fceb-d81e-498c-a121-19a3292f7f7f&id=70b367b-fc69-490b-9e8b-cf74db9541&id=d6f8889-977c-4769-88eb-b0f7013c1ee8&id=d3b10d5-777d-4b44-e978-e
55abe69370f
2025-10-26T18:38:15.317811Z INFO /omniioth/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/encrypted-profiles/
71d089410e-24a9-4a10-b21b-fe3da28dddb8&id=ef5fceb-d81e-498c-a121-19a3292f7f7f&id=70b367b-fc69-490b-9e8b-cf74db9541&id=d6f8889-977c-4769-88eb-b0f7013c1ee8&id=d3b10d5-777d-4b44-e978-e
55abe69370f
2025-10-26T18:38:22.358306Z INFO /omniioth/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/encrypted-profiles/
71d089410e-24a9-4a10-b21b-fe3da28dddb8&id=ef5fceb-d81e-498c-a121-19a3292f7f7f&id=70b367b-fc69-490b-9e8b-cf74db9541&id=d6f8889-977c-4769-88eb-b0f7013c1ee8&id=d3b10d5-777d-4b44-e978-e
55abe69370f
2025-10-26T18:38:24.274589Z INFO /omniioth/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/devices/923f1655-aa0
7-4d8c-8220-528ab3ac9b98/
2025-10-26T18:38:24.761937Z INFO /omniioth/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.ibm.com/api/manufacturers/f523e2
c4-6841-4585-a5d1-2ae3da08609a/
2025-10-26T18:38:25.527161Z INFO edge-gateway/src/lib.rs:139: Connecting to modbus service at address '192.168.1.3:10502'
2025-10-26T18:38:25.530553Z INFO edge-gateway/src/lib.rs:152: Connected to modbus service at address '192.168.1.3:10502'
2025-10-26T18:38:25.535800Z INFO edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.
ibm.com:443'
2025-10-26T18:38:27.089117Z INFO edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eb77bb.am1.techzone.i
bm.com:443'
```

8. Verify the selected Lenze data in Monitor. Click to open the **XX_Lenze_i550_01** device.

Device types

Search

DVC_DB_TP
● Analysis off

Lenze_DT_TP
● Analysis off

LENZE_VFD
● Analysis off

TP_DVC_TYPE
● Analysis off

Lenze_DT_TP

Search

Device

Device name (Alias)

Description

TP_Lenze_i550_01

9. Navigate to **Recent event** and wait for a minute until the first message is coming through.

Device type / TP_Lenze_i550_01

Device

● Analysis off

Manage device type

Data Overview Meters mapping Metrics Calculated metrics **Recent event** Dashboards Dimensions

Event	Value	Format	Last received	
status	{"timestamp":"2025-10-26T18:48:13.369227Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:12.344413Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:11.315146Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:10.289967Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:09.261415Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:08.233504Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:07.214045Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago
status	{"timestamp":"2025-10-26T18:48:06.190359Z","ControlCardTemperature_degreeCelsius":23.2000...	View payload	json	a few seconds ago



Note- If the recent event is not showing any payload. Then go to the terminal where you run the docker command and copy the MQTT service append `https` before the URL and launch in the browser like:

```
2025-10-26T18:38:25.535880Z: INFO: edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443'
2025-10-26T18:38:27.009117Z: INFO: edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443'
```

<https://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443>

and accept the certificate. Refresh the device you will see the payload.

10. Click on **View payload** and see the data points being send to the Event name status.



11. To see the data in the metrics. Go to Data -> Control Card Temperature -> Data Table.

ControlCardTemperature_degreeCelsius (Metric) Last updated: 27/10/2025 00:43:28

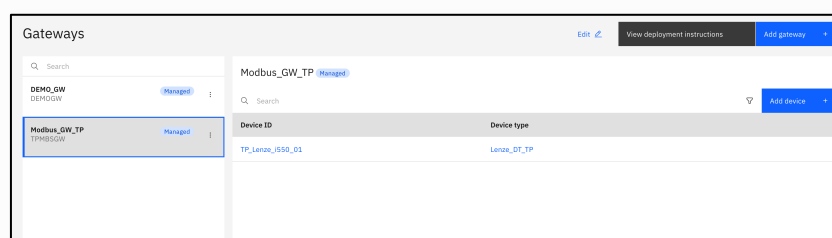
Unit: degree Celsius

Timestamp	Device ID	ControlCardTemperature_degreeCelsius
27/10/2025 00:43:28	TP_Lenze_i550_01	23.8
27/10/2025 00:43:27	TP_Lenze_i550_01	23.8
27/10/2025 00:43:26	TP_Lenze_i550_01	23.8
27/10/2025 00:43:25	TP_Lenze_i550_01	23.8
27/10/2025 00:43:24	TP_Lenze_i550_01	23.8
27/10/2025 00:43:23	TP_Lenze_i550_01	23.8
27/10/2025 00:43:22	TP_Lenze_i550_01	23.8
27/10/2025 00:43:21	TP_Lenze_i550_01	23.8
27/10/2025 00:43:20	TP_Lenze_i550_01	23.8
27/10/2025 00:43:19	TP_Lenze_i550_01	23.8

Exercise 5. Add another Industrial Device

In this Exercise you will learn how to add a second industrial device to the Managed Gateway.

1. Open the Gateway in Monitor that you have created.



2. Click on **Add device**.

Modbus_GW_TP

Add device to TPMSGW

Configuration
Managed

Managed gateways support only devices from the library.

Use device library ✓
Choose a device from a library of preconfigured, manufacturer-specific devices. [Learn more about the library](#)

Add custom device
Configure your own device. Specify custom device details and metrics. [Learn more about custom devices](#)

Cancel Continue

3. Select Manufacturer: **Lenze** and product: **i550** and click on Next.

Gateway / Add device

Select device
Select a manufacturer and then search for a device in the library.

Manufacturer: Lenze

Search

	Product name	Product series	Product type
☐	9400 Highline	E94	Variable Frequency Drives
☐	BaseLine D	B400	Variable Frequency Drives
☐	i510	i500	Variable Frequency Drives
☒	i550	i500	Variable Frequency Drives
☐	i550 protec	i500	Variable Frequency Drives
☐	IP65	SMVector	Variable Frequency Drives
☐	Motec	B400	Variable Frequency Drives
☐	NEMA-1	SMV	Variable Frequency Drives
☐	NEMA-4X	SMV	Variable Frequency Drives

Cancel Next

4. Now it is time to use the IP address of the simulator and combine it with port number 20502 separated by : , like **192.168.1.64:20502**. Click on **Next**.

Gateway / Add device

Configure protocol
Select a protocol for the device and then configure the protocol-specific values.

Protocol: Modbus TCP

IP address: 192.168.1.64:20502

Server or unit ID: 1

Cancel Previous Next

5. Enter the Device ID and Device Type:

- Device ID: **XX_Lenze_i550_02**
- Device type: **Lenze_DT1_XX**

The screenshot shows the 'Add device' configuration page with a progress bar on the left indicating the current step is 'Configure device'. The main area is titled 'Configure device' and contains two input fields: 'Device ID' with the value 'TP_Lenze_i550_02' and 'Device type' with a dropdown menu showing 'Lenze_DT1_TP'. A 'Product type' field on the right shows 'Variable Frequency Drives'. At the bottom, there are 'Cancel', 'Previous', and 'Next' buttons.

6. Select these three metrics and click on Next.

The screenshot shows the 'Add device' configuration page with a progress bar on the left indicating the current step is 'Select metrics'. The main area is titled 'Select metrics' and contains a table of metrics. Three metrics are selected: 'Motor Current', 'Motor Torque', and 'Motor Voltage'. The table has columns for 'Metric', 'Unit', 'Data frequency (ms)', and 'Change threshold (%)'. At the bottom, there are 'Cancel', 'Previous', and 'Next' buttons.

Metric	Unit	Data frequency (ms)	Change threshold (%)
☐ Control Card Temperature	degree Celsius	1000	On every change
☐ DC Bus Voltage	volt	1000	On every change
☐ Frequency Command	hertz	1000	On every change
☒ Motor Current	ampere	1000	On every change
☒ Motor Torque	percent	1000	On every change
☒ Motor Voltage	volt	1000	On every change
☐ Output Frequency	hertz	1000	On every change
☐ Torque Command	percent	1000	On every change
☐ Supply Voltage	volt	1000	On every change

7. Review the Summary and click on Save.

The screenshot shows the 'Add device' configuration page with a progress bar on the left indicating the current step is 'Summary'. The main area is titled 'Summary' and contains a summary of the device configuration details. At the bottom, there are 'Cancel', 'Previous', and 'Save' buttons.

Device Details	
Manufacturer	Lenze
Product name	i550
Gateway ID	Modbus_GW_TP
Gateway name	TPMB5GW
Device ID	TP_Lenze_i550_02
Device type	Lenze_DT1_TP
Protocol Details	
Protocol	modbus-tcp
IP/Port/Command	192.168.1.5:20502/1

8. Both the devices will be reflecting under the gateway.

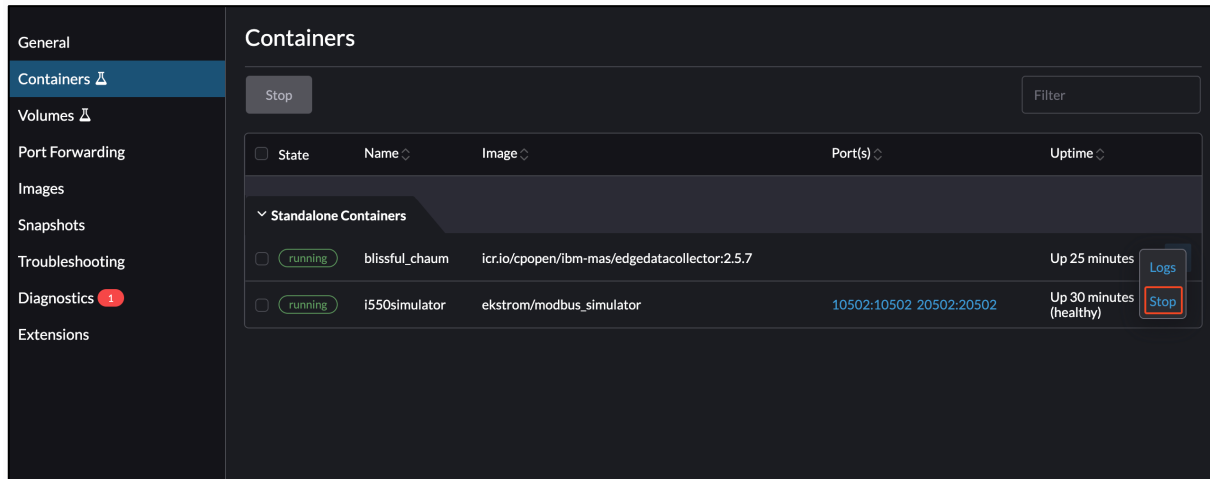


And then run this command **docker kill <container id>**

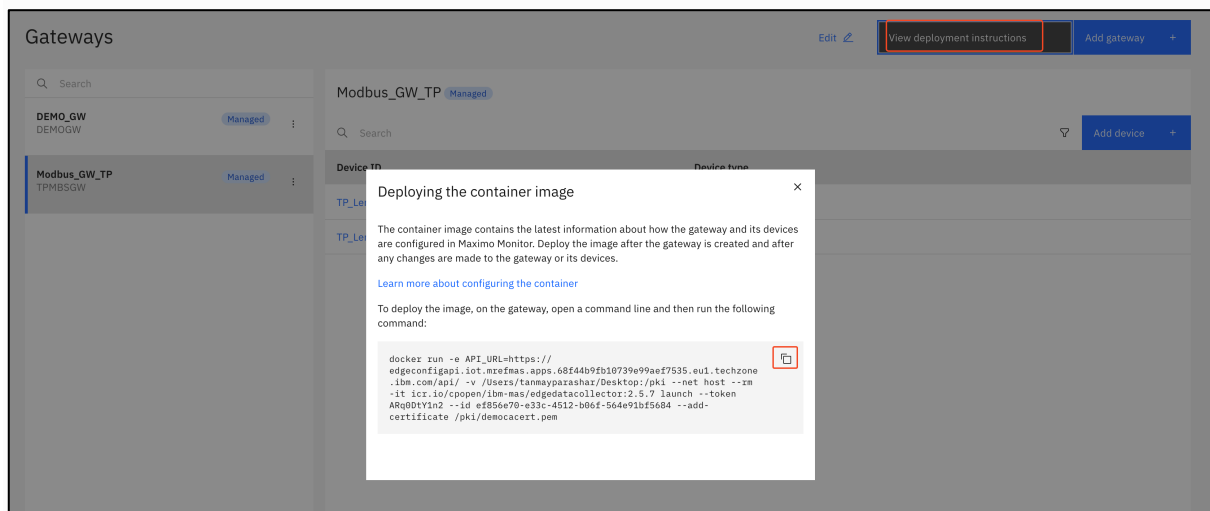
In this exercise the command would be: **docker kill 156c9fc21cb8**

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker kill 156c9fc21cb8
```

OR You can directly kill the container from Rancher Desktop Application by clicking on Stop.



5. Navigate back to your Managed Gateway in Monitor and press the **View deployment instructions**. Click on the docker command to copy it to the clipboard.



6. Get back to the terminal and then paste the docker command line from the clipboard. Click enter to execute it, and you should see something similar to the following.


```
tanmayparashar@Tanmays-MacBook-Pro-2: ~ % docker run -e API_URL=https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/ --v /Users/tanmayparashar/Desktop/pk
1 --net host --rm -it icr.io/cpopen/lib-mas/edgedatacollector:2.5.7 launch --token ARQ0DYIn2 --id ef85de78-e33c-4512-b86f-564e91bf5684 --add-certificate /pk1/democacert.pem
WARNNG: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-27T18:38:14.734948Z INFO: edge-cli/src/lib.rs:186: Arguments: launch --token ARQ0DYIn2 --id ef85de78-e33c-4512-b86f-564e91bf5684 --add-certificate /pk1/democacert.pem
2025-10-27T18:38:14.737955Z INFO: edge-cli/src/lib.rs:108: Version: 2.5.7
2025-10-27T18:38:14.738028Z INFO: edge-cli/src/lib.rs:109: Git sha: cab0ca6f2bf01743db0f93cd83ab9763bc4668
2025-10-27T18:38:14.738092Z INFO: edge-cli/src/lib.rs:110: Branch: master
2025-10-27T18:38:14.738094Z INFO: edge-cli/src/lib.rs:111: Build number: 748
2025-10-27T18:38:14.738129Z INFO: edge-cli/src/lib.rs:112: Build ID: 41827330
2025-10-27T18:38:14.864945Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/installations/ef85de
78-e33c-4512-b86f-564e91bf5684/
2025-10-27T18:38:15.978211Z INFO: /omniot/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/encrypted-profiles/
71d56c37254a-752f-445b-af2e-84b32d9cbbcd&id=08904184-2a49-4b10-b21b-fe3a28bddd&id=94b0dac3-7d22-439b-8c2e-eeb9527cc25a&id=afc5fceb-d81e-498c-a121-1963292f77f7&id=0631f173-aded-4172-b9f1-b
11553f94c4c
2025-10-27T18:38:16.986131Z INFO: /omniot/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/encrypted-profiles/
71d56c37254a-752f-445b-af2e-84b32d9cbbcd&id=08904184-2a49-4b10-b21b-fe3a28bddd&id=94b0dac3-7d22-439b-8c2e-eeb9527cc25a&id=afc5fceb-d81e-498c-a121-1963292f77f7&id=0631f173-aded-4172-b9f1-b
11553f94c4c
2025-10-27T18:38:17.915688Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/units/
2025-10-27T18:38:18.281578Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/devices/923f1655-aa9
7-4d8c-822b-52ab3ac3b985/
2025-10-27T18:38:18.629908Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/manufacturers/f523e2
04-4841-4585-e355-2ac3dc98e99a/
2025-10-27T18:38:19.332888Z INFO: edge-gateway/src/lib.rs:139: Connecting to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.334331Z INFO: edge-gateway/src/lib.rs:152: Connected to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.337974Z INFO: edge-gateway/src/lib.rs:139: Connecting to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.338383Z INFO: edge-gateway/src/lib.rs:152: Connected to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.342028Z INFO: edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://mrefmas.messaging.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.
ibm.com:443'
2025-10-27T18:38:20.795848Z INFO: edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://mrefmas.messaging.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.i
bm.com:443'
```

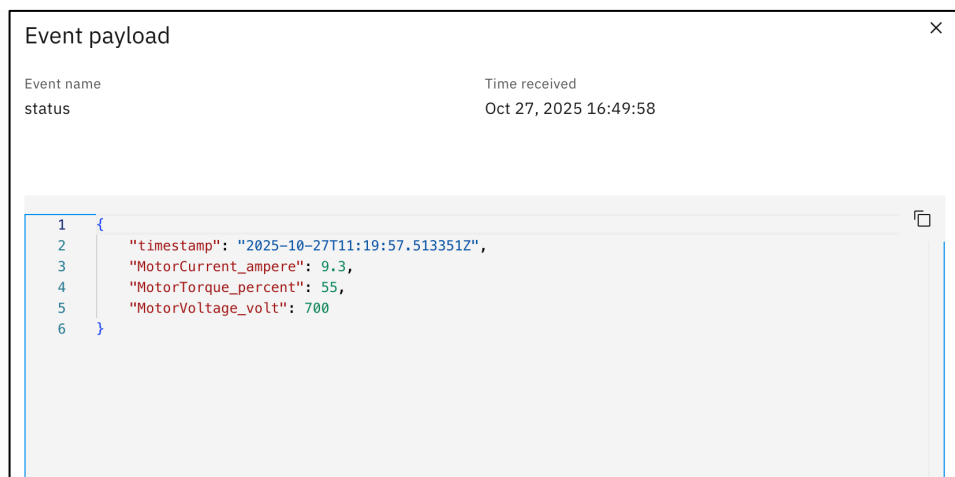


NOTE: In this step, the IP address is set to localhost. This situation occurs when the network does not have a fixed IP address. If you notice that your network configuration or IP address (as seen using the **ifconfig en0** command) changes frequently, switch to your local IP address (commonly **127.0.0.1**, though it may vary depending on your system). Then, open the device you have created in your setup and update its IP address accordingly.

7. Navigate to **Recent event** for the second device and wait for a minute until the first message is coming through.

Gateways / TP_Lenze_i550_02				
Device				
Analysis off Manage device type				
Data	Overview	Meters mapping	Calculated metrics	Recent event
Dashboards Dimensions				
Event	Value	Format	Last received	
status	["timestamp":"2025-10-27T11:17:19.404207Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:18.372952Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:17.360682Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:16.342997Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:15.322411Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:14.297568Z","MotorCurrent_ampere":9.5,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:13.270750Z","MotorCurrent_ampere":9.5,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:12.260398Z","MotorCurrent_ampere":9.5,"MotorTorque_percent... View payload	json	a few seconds ago	

8. Click on **View payload** and see the data points being send to the Event name status.

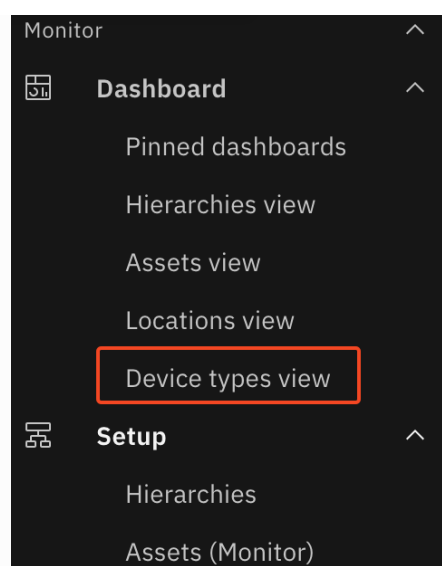


9. To see the data in the metrics. **Go to Data -> Motor Current -> Data Table.**

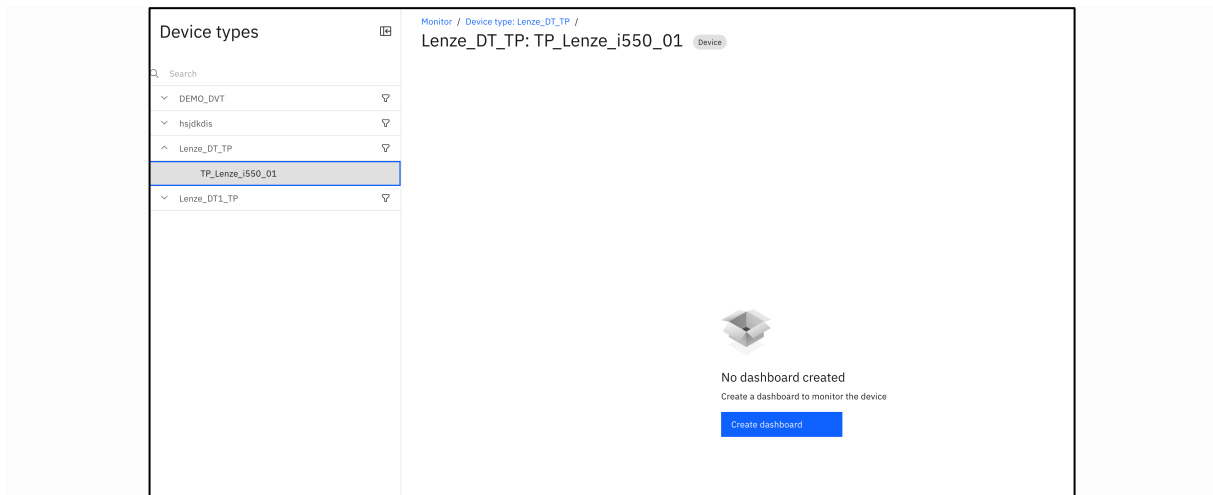
The screenshot shows the "Data" view for the "MotorCurrent_ampere" metric. The left sidebar lists various data items, with "Motor Current" highlighted. The main area displays a table of data points.

Timestamp	Device ID	MotorCurrent_ampere
10/27/2025 16:51:01	TP_Lenze_i550_02	9.3
10/27/2025 16:51:00	TP_Lenze_i550_02	9.3
10/27/2025 16:50:59	TP_Lenze_i550_02	9.3
10/27/2025 16:50:58	TP_Lenze_i550_02	9.3
10/27/2025 16:50:57	TP_Lenze_i550_02	9.3
10/27/2025 16:50:56	TP_Lenze_i550_02	9.3
10/27/2025 16:50:55	TP_Lenze_i550_02	9.3
10/27/2025 16:50:54	TP_Lenze_i550_02	9.3
10/27/2025 16:50:53	TP_Lenze_i550_02	9.3

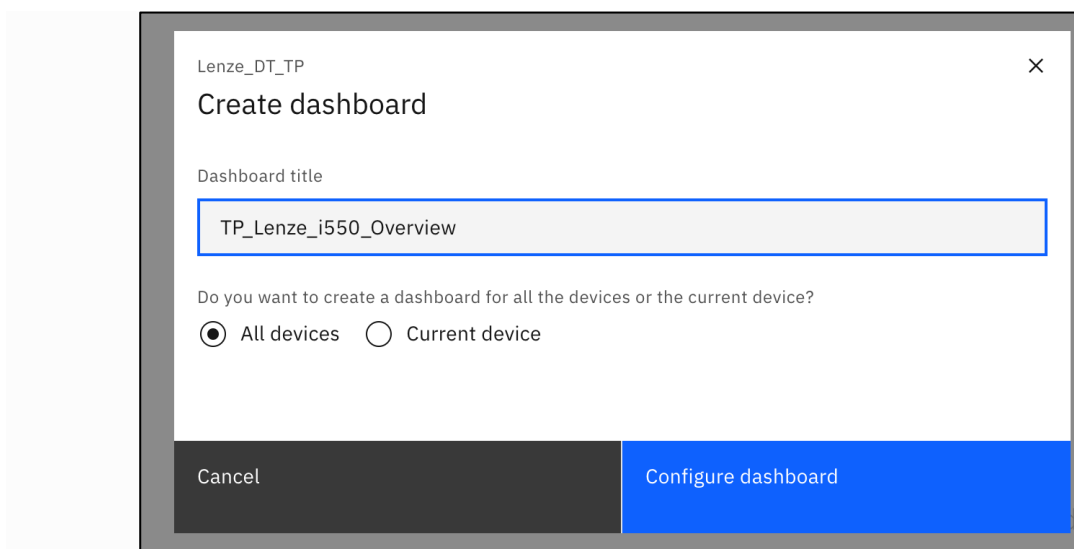
10. Navigate to **Dashboard -> Device Type View** in Monitor.



11. Select the **XX_Lenze_i550_01** device and then click on **Create Dashboard**

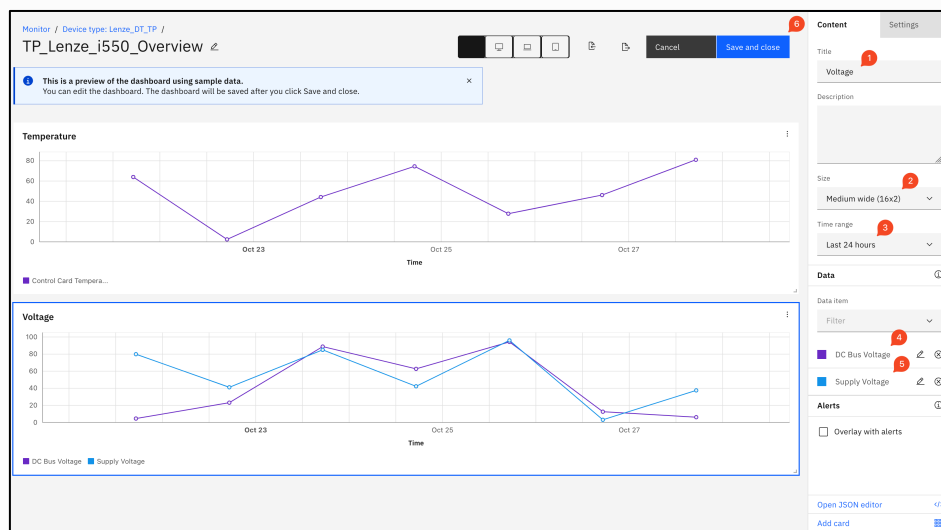
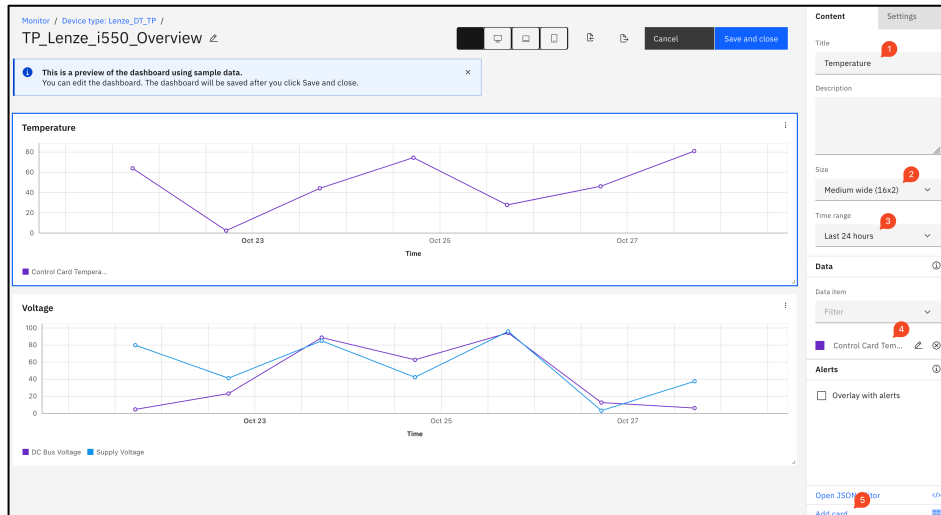


12. Give your new dashboard a name and press **Configure dashboard**.
Title: **XX_Lenze_i550_Overview**

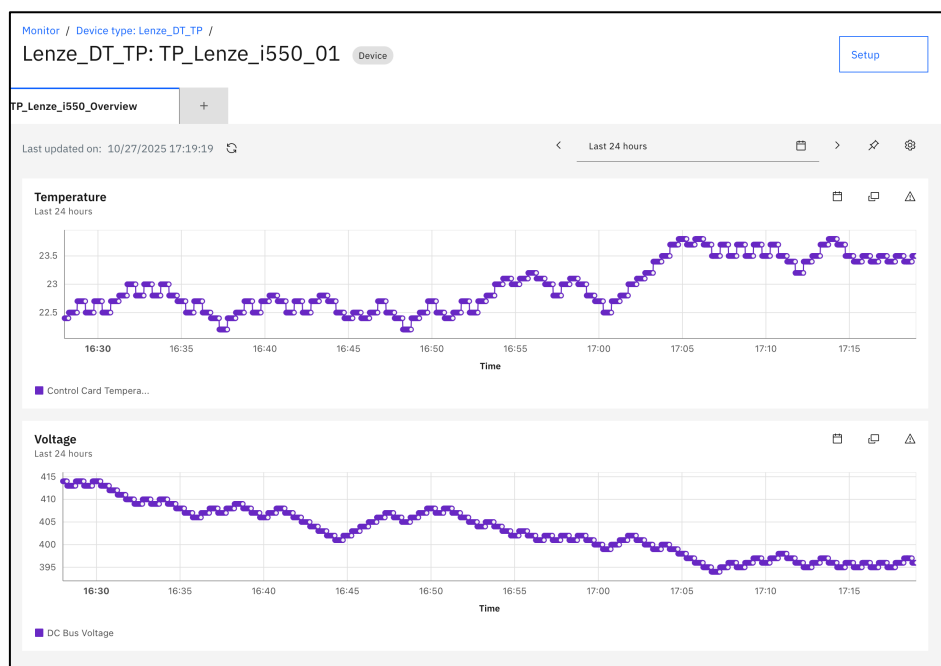


13. You will now see an empty dashboard configuration. Add a Time series line card by clicking on it.

- Locate the Control Card Temperature in the Data item dropdown and change the Time range to Last 24 hours.
- Give the card a title, like Temperature [°C] and drag the lower right corner to resize the card to full width.
- Once that is in place, then Add another Time series line card.
- Configure the second Time series line card according to the screen shot below and press Save and Close.



14. The dashboard for the first device will look like this.



15. Similarly you can configure for second device for different metrics.

